

Interpretability Theory Part 2: Sparse Autoencoders

LIN 313 Language and Computers
UT Austin Fall 2025

Admin (12/1)

- **Important: Finish annotation assignment if you have not already!!! We need the complete dataset for a good classifier!**
- HW 4 to be returned Wednesday
- Reading for Wednesday: [Published on Mar 27, 2024 DOI Who Do We Become When We Talk to Machines?](#)
- Final Exam
 - Thurs 12/11 1pm-3pm RLP 1.104
 - 2 hours
 - one 2-sided 8.5x11 **handwritten** notesheet
- Exam Prep
 - focus on new post-midterm topics but there will be some questions covering earlier topics (n-grams, naive bayes, perceptrons, personas)
 - review homeworks, midterm (midterm solutions available to view in office hours or by appointment)
 - Practice questions distributed Wednesday 12/3
 - Optional review sessions
- Final project report :
 - test and evaluate classifier
 - brief (~1 page) write-up results, analysis of the category, and recommendations for future experiments.
 - optional: build your own version of a classifier application instead.
- Option: work on report in class or AI & Workplace Panel:
<https://polls.la.utexas.edu/course/5422/teacher#/polls/17%20-%20Interpretability>

Key Post-Midterm Topics

- Word Embeddings (Word2Vec) - vectors distributional semantics, cosine similarity, analogies, bias in embeddings
- Contextual Embeddings & Polysemy - BERT, Context Atlas, affordances
- Attention & Transformers - attention mechanisms, translation, alignment
- RLHF (Reinforcement Learning from Human Feedback)
- Can Machines Think? / Do LLMs Understand Language? - Turing Test, philosophical questions
- Interpretability - mechanistic interpretability
- Chatbots & Gricean Maxims - conversational AI, pragmatics

Key Pre-Midterm Topics

- What makes language unique?
- Ambiguity in Language - syntactic and semantic
- Elements of conversation
 - Gricean Maxims
 - Jakobsonian speech functions
- Symbolic AI (GOFAI)
- Naive Bayes for text classification
 - Conditional probability
- N-gram language models
 - Conditional probability again!
- The Perceptron
- Inter-annotator agreement (IAA)
- Chatbot Personas

Overview for Today

- Results of group project annotation
- Sparse Autoencoders

Classifier Annotation Results

Calculating Agreement

- Average pairwise Cohen's kappa
 - for each pair of annotators, calculate Cohen's kappa
 - then average the values
- Krippendorff's α (alpha)
 - handles missing data and multiple annotators more robustly
 - just for comparison, don't worry about the details
- Consensus
 - Which of the examples did all annotators agree on?

Classifier Annotation Results - Brainrot v Thoughtful

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IAA FOR: Group 1 - Brainrot vs Thoughtful (Binary)
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```
=====
Rows with annotations: 896
Number of annotators: 9
Number of items: 600
jbr2988 vs mey374:  $\kappa = 0.421$  (n=100)
jl86274 vs jls22669:  $\kappa = -0.093$  (n=99)
jl86274 vs ss225978:  $\kappa = 0.047$  (n=97)
jls22669 vs ss225978:  $\kappa = 0.000$  (n=96)
```

Mean pairwise κ : 0.094

```
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=====
IAA FOR: Group 2 - Brainrot vs Thoughtful (Ternary)
=====
```

```
=====
Rows with annotations: 1153
Number of annotators: 12
Number of items: 600
djs5342 vs sd45682:  $\kappa = 0.418$  (n=100)
djs5342 vs sl59226:  $\kappa = 0.228$  (n=100)
gsc685 vs lgm2244:  $\kappa = -0.082$  (n=54)
hsb766 vs nq732:  $\kappa = 0.318$  (n=100)
rmp3599 vs rrt764:  $\kappa = 0.066$  (n=100)
sd45682 vs sl59226:  $\kappa = 0.085$  (n=100)
teg883 vs ylm275:  $\kappa = 0.162$  (n=99)
```

Mean pairwise κ : 0.171

Classifier Annotation Results - Red Flag vs Green Flag

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IAA FOR: Group 1 - Red Flag vs Green Flag (Binary)
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Rows with annotations: 899

Number of annotators: 9

Number of items: 600

j186274 vs jls22669: $\kappa = 0.140$ (n=100)

j186274 vs ss225978: $\kappa = 0.378$ (n=99)

jls22669 vs ss225978: $\kappa = 0.149$ (n=99)

kab5987 vs sc74299: $\kappa = 0.234$ (n=100)

Mean pairwise κ : 0.225

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IAA FOR: Group 2 - Red Flag vs Green Flag (Ternary)
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```

```
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```

Rows with annotations: 1098

Number of annotators: 11

Number of items: 600

djs5342 vs sd45682: $\kappa = 0.248$ (n=100)

djs5342 vs sl59226: $\kappa = 0.521$ (n=100)

hsb766 vs nq732: $\kappa = 0.165$ (n=98)

rmp3599 vs rrt764: $\kappa = 0.173$ (n=100)

sd45682 vs sl59226: $\kappa = 0.186$ (n=100)

teg883 vs ylm275: $\kappa = 0.432$ (n=100)

Mean pairwise κ : 0.287

Cohen's Kappa - Summary

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SUMMARY OF INTERANNOTATOR AGREEMENT RESULTS					
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	Group Mean Kappa	N Annotators	N Items	N Pairwise Comparisons	
Group 2 - Brainrot vs Thoughtful	0.171	12	600		7
Group 2 - Red Flag vs Green Flag	0.287	11	600		6
Group 1 - Brainrot vs Thoughtful	0.094	9	600		4
Group 1 - Red Flag vs Green Flag	0.225	9	600		4

```
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KAPPA INTERPRETATION:

$\kappa < 0.00$: Poor agreement

$0.00 \leq \kappa \leq 0.20$: Slight agreement

$0.21 \leq \kappa \leq 0.40$: Fair agreement

$0.41 \leq \kappa \leq 0.60$: Moderate agreement

$0.61 \leq \kappa \leq 0.80$: Substantial agreement

$0.81 \leq \kappa \leq 1.00$: Almost perfect agreement

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Krippendorff's Alpha

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COMPARISON: COHEN'S KAPPA vs KRIPPENDORFF'S ALPHA
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	Group Mean Cohen κ	Krippendorff α	Difference ($\alpha - \kappa$)
Group 2 - Brainrot vs Thoughtful	0.171	0.195	0.025
Group 2 - Red Flag vs Green Flag	0.287	0.324	0.036
Group 1 - Brainrot vs Thoughtful	0.094	0.165	0.071
Group 1 - Red Flag vs Green Flag	0.225	0.238	0.013

```
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NOTES ON DIFFERENCES:
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- Krippendorff's α handles missing data and multiple annotators more robustly
 - Cohen's κ is calculated pairwise, then averaged
 - Both measure nominal agreement, but α accounts for chance differently
 - Generally expect α to be slightly lower than mean pairwise κ
- ```
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# Consensus

## =====

### COMPARISON: ALL ITEMS vs CONSENSUS ITEMS

## =====

|                                  | Group | Consensus Rate | Consensus Items | Total Items |
|----------------------------------|-------|----------------|-----------------|-------------|
| Group 2 - Brainrot vs Thoughtful |       | 45.0%          | 204/453         | 600         |
| Group 2 - Red Flag vs Green Flag |       | 51.3%          | 204/398         | 600         |
| Group 1 - Brainrot vs Thoughtful |       | 48.5%          | 97/200          | 600         |
| Group 1 - Red Flag vs Green Flag |       | 52.5%          | 105/200         | 600         |

### CONSENSUS FILTERING EFFECTS:

- Consensus items have 100% annotator agreement
- Higher agreement scores on consensus subset indicate:
  - Some items are inherently easier to classify
  - Disagreement may cluster on ambiguous cases
  - Training could focus on difficult/ambiguous items

### IMPLICATIONS:

- High consensus rates (>50%) suggest clear category boundaries
- Low consensus rates (<30%) may indicate:
  - Unclear annotation guidelines
  - Inherently subjective categories
  - Need for additional annotator training
- Large improvements in  $\kappa/\alpha$  on consensus items suggest:
  - Disagreement is not random but systematic
  - Some items have clearer correct answers

# Consensus

```
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COMPARISON: ALL ITEMS vs CONSENSUS ITEMS
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## RECOMMENDATIONS:

- Group 2 - Brainrot vs Thoughtful: Moderate consensus (45.0%) - consider refining guidelines
- Group 2 - Red Flag vs Green Flag: Good consensus rate (51.3%) - annotation task is well-defined
- Group 1 - Brainrot vs Thoughtful: Moderate consensus (48.5%) - consider refining guidelines
- Group 1 - Red Flag vs Green Flag: Good consensus rate (52.5%) - annotation task is well-defined

```
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```

- For red flag/green flag, IAA is only "fair" (kappa  $\sim .22 - .28$  ), but consensus is solid ( $\sim 51\text{-}52\%$ )
- How can consensus be high but inter-annotator agreement be low?

# Clear Cases vs. Ambiguous Cases

## The Core Issue:

- Many items are actually easy to classify - annotators largely agree (high consensus rate)
- Remaining items are highly problematic - annotators disagree substantially (drags down IAA)

## Concrete Example:

- If you have 100 items:
  - 70 items: 90% annotator agreement (clear consensus)
  - 30 items: 20% annotator agreement (total chaos)

What could this mean about our data / task?

# Clear cases vs. Ambiguous Cases

## 1. Task Has two types of items

- Type A: "Obviously red flag" / "Obviously green flag" → Clear, unambiguous cases
- Type B: "Could go either way" → Genuinely difficult/ambiguous cases

## 2. Guidelines Work... Sometimes

- Guidelines are sufficient for clear cases, insufficient for edge cases
- Solution: need more specific rules for handling ambiguous situations

## 3. Systematic vs Random Disagreement

- Large improvements in  $\kappa/\alpha$  on consensus items suggest disagreement is not random but systematic
- Some items have clearer "correct" answers than others



# Practical Implications

**"Easy cases are easy, hard cases are really hard"** - This is actually a common and manageable situation that tells you exactly where to focus improvement efforts!

- **Good news:** We have substantial reliable data (consensus items)
- **Challenge:** systematic disagreement on remaining items - **is this because the concept is too subjective? or to do with task design.**

For training this classifier:

- Use consensus items for initial model training

For future work:

1. Analyze disagreement items - what makes them difficult?
2. Improve guidelines - add specific rules for edge cases
3. Targeted training - focus on ambiguous cases

# Building a Gold Standard

Low IAA is actually fairly standard for NLP Tasks. Many NLP tasks, like 'toxic language detection', 'sentiment analysis', are inherently subjective. But there is still interest in developing predictive models for them.

Common practice is to limit the dataset to unambiguous cases - cases where most annotators agree.

Our solution: limit to data points where all of our anotators agree.

# A classifier app

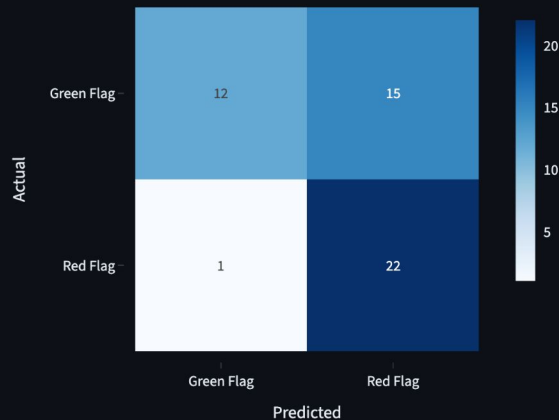
## Evaluation Results

### Classification Metrics

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Green Flag   | 0.923     | 0.444  | 0.6      | 27      |
| Red Flag     | 0.595     | 0.957  | 0.733    | 23      |
| accuracy     | 0.68      | 0.68   | 0.68     | 0.68    |
| macro avg    | 0.759     | 0.7    | 0.667    | 50      |
| weighted avg | 0.772     | 0.68   | 0.661    | 50      |



### Confusion Matrix



# LLM Interpretability

We know the model is about to become doubtful.



But there is no single neuron corresponding to 'doubt'. How do we isolate the parts of the LLM that process the concept of 'doubt'?

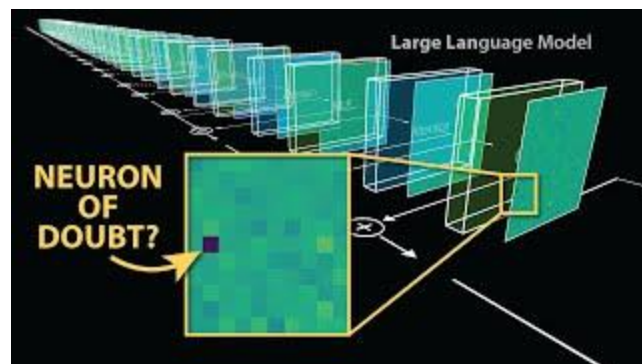
# How does Golden Gate Claude Work? Sparse Autoencoders

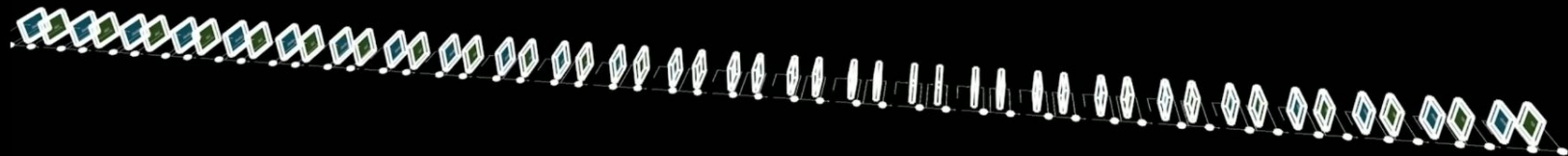
SAEs are like the die (the cutting disk) in this pasta extrusion machine.

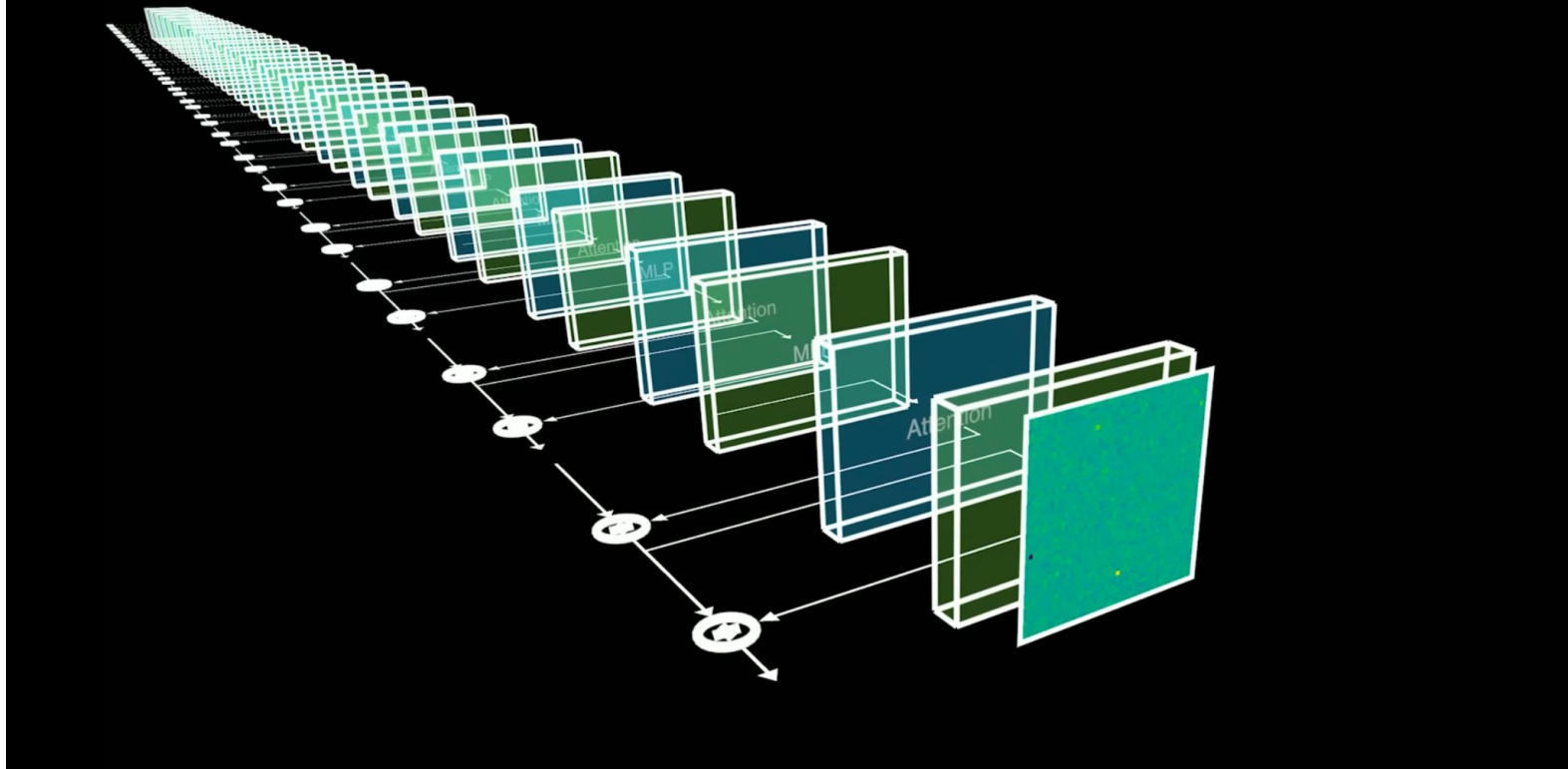
Imagine dough glowing through a tube. We put an obstacle in the way of the flow, and force the dough to become the pasta that it has been all along.

Then we squish the pasta back into dough











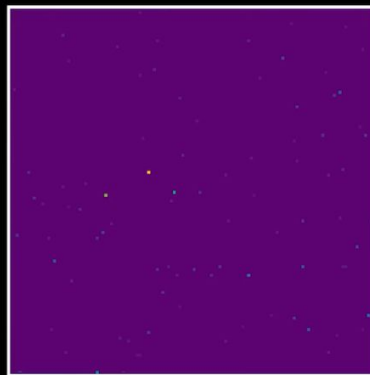


Force most terms to 0

Reshape

$$[ 0 \quad f_1 \quad 0 \quad \dots \quad 0 ]$$

SPARSE CONCEPT/FEATURE VECTOR



# Untangling 'doubt'

$$\begin{bmatrix} a_0 & a_1 & a_2 & \dots & a_{2303} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \\ \dots \\ w_{2303} \end{bmatrix}$$

NEURON OUTPUTS (ACTIVATIONS)

WEIGHTING VECTOR

# Untangling 'doubt'

$$\begin{array}{c} \overbrace{[a_0 \quad a_1 \quad a_2 \quad \dots \quad a_{2303}]}^{2,304} \\ \text{NEURON OUTPUTS (ACTIVATIONS)} \end{array} \left[ \begin{array}{c} w_0 \\ w_1 \\ w_2 \\ \dots \\ w_{2303} \end{array} \right] \left. \vphantom{\begin{array}{c} w_0 \\ w_1 \\ w_2 \\ \dots \\ w_{2303} \end{array}} \right\} 2,304$$

WEIGHTING VECTOR

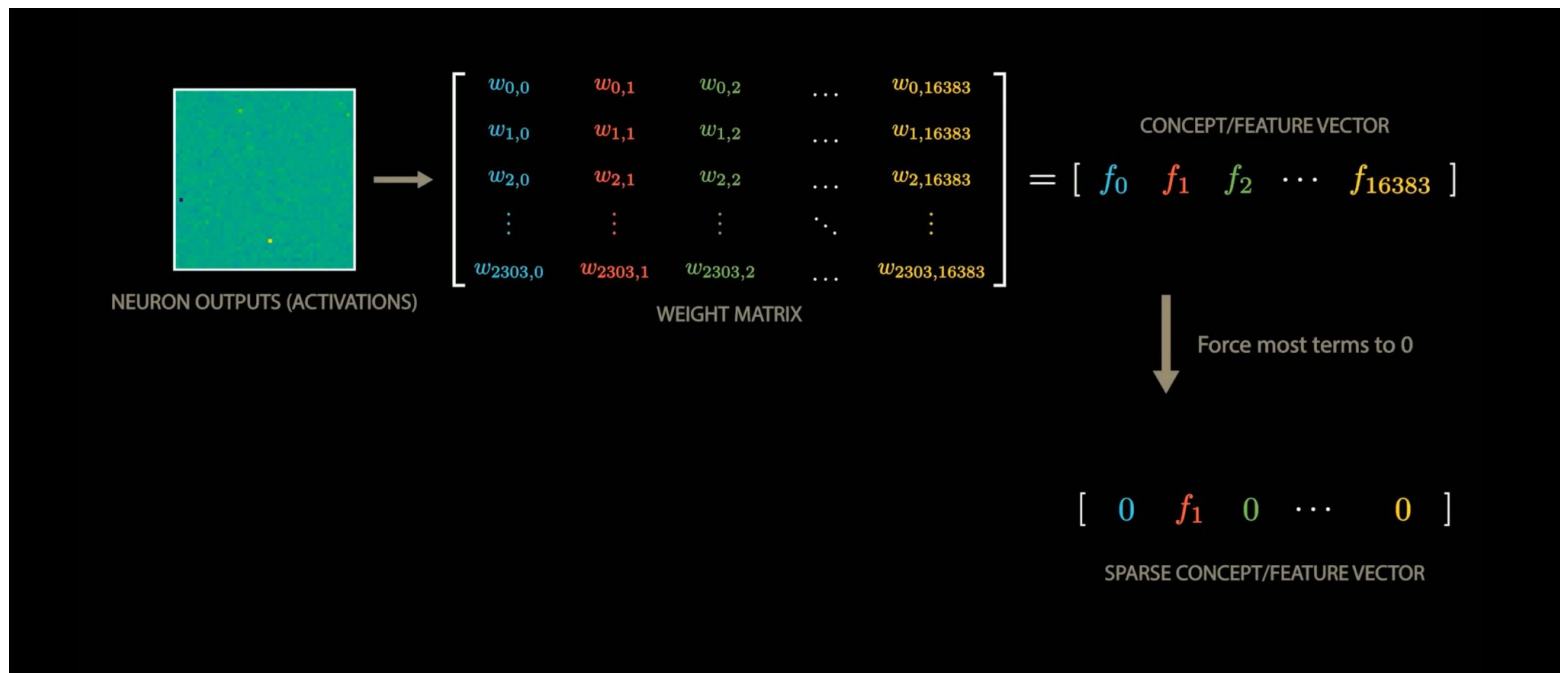
# Untangling 'doubt'

$$\begin{array}{c} \left[ \begin{array}{ccccccc} a_0 & a_1 & a_2 & \dots & a_{2303} \end{array} \right] \\ \text{NEURON OUTPUTS (ACTIVATIONS)} \end{array} \left[ \begin{array}{c} w_0 \\ w_1 \\ w_2 \\ \dots \\ w_{2303} \end{array} \right] = a_0 w_0 + a_1 w_1 + a_2 w_2 + \dots + a_{2303} w_{2303}$$

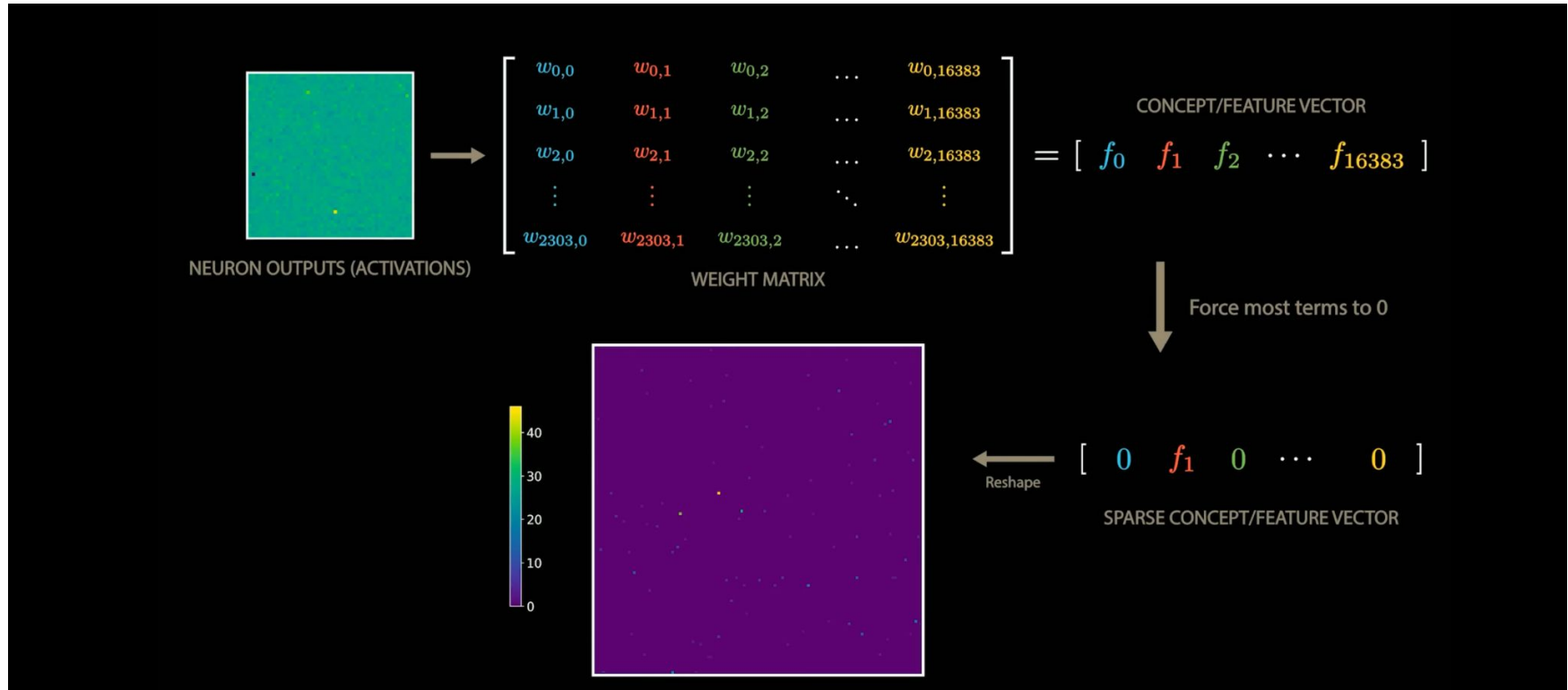
WEIGHTING VECTOR

= STRENGTH OF CONCEPT

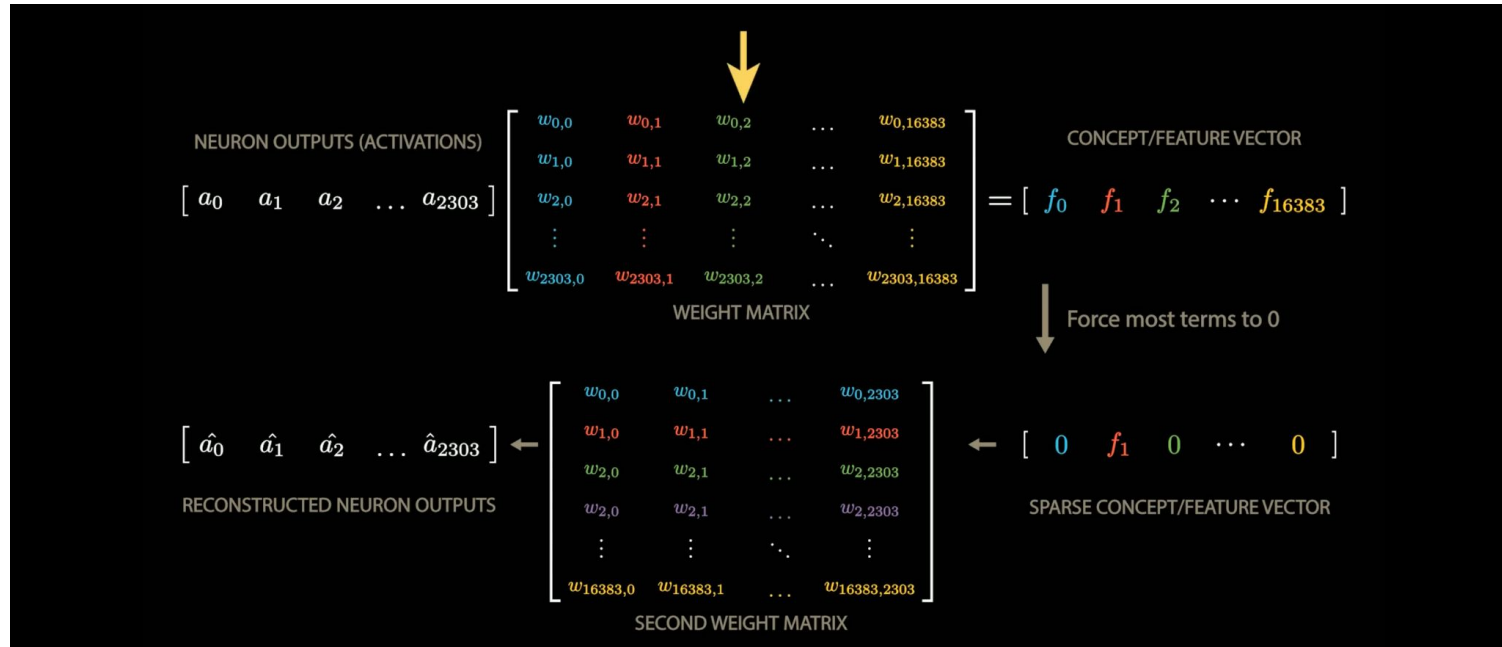
Squeeze the mixed-up dough into the pasta it was inside all along



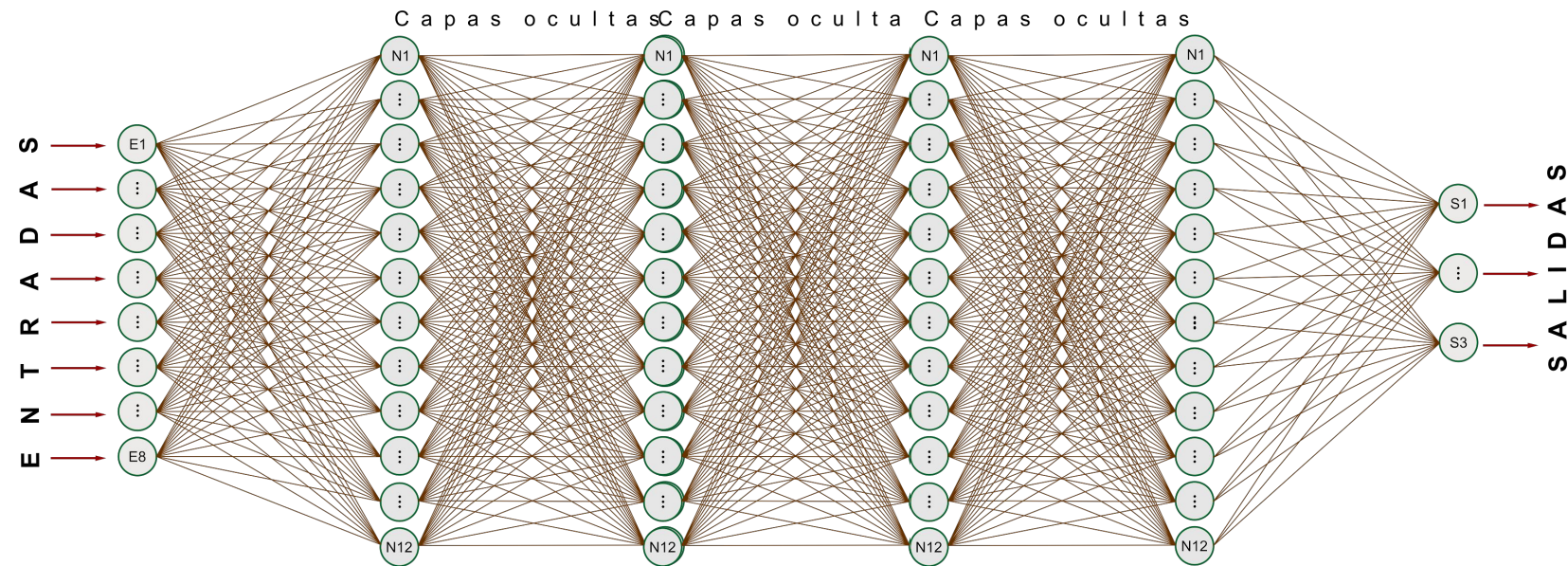
# Look at the pasta shapes this input makes



# Squish the pasta shapes back into the same mixed up tangle

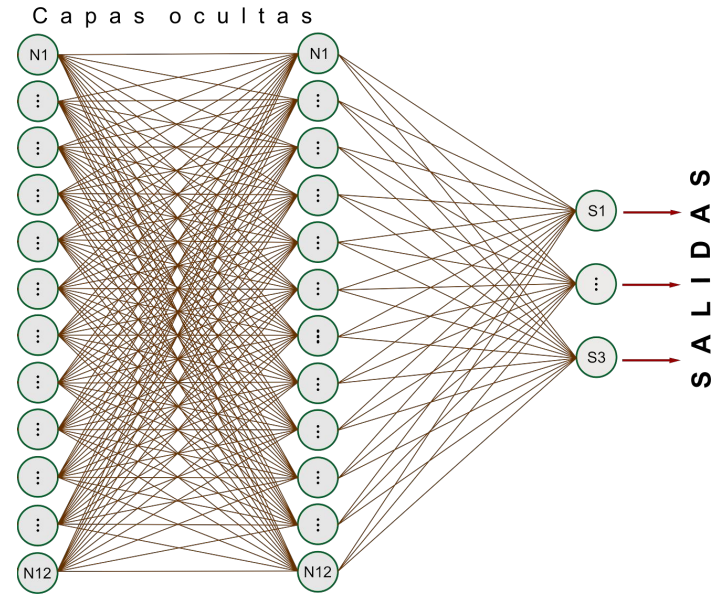
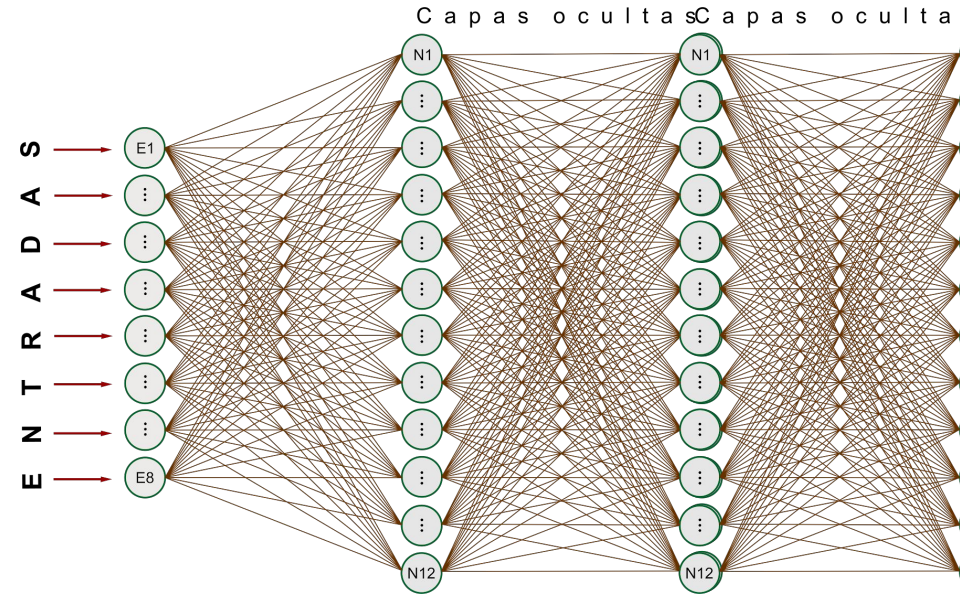


# Sparse Autoencoders: surgery and spying on LLMs

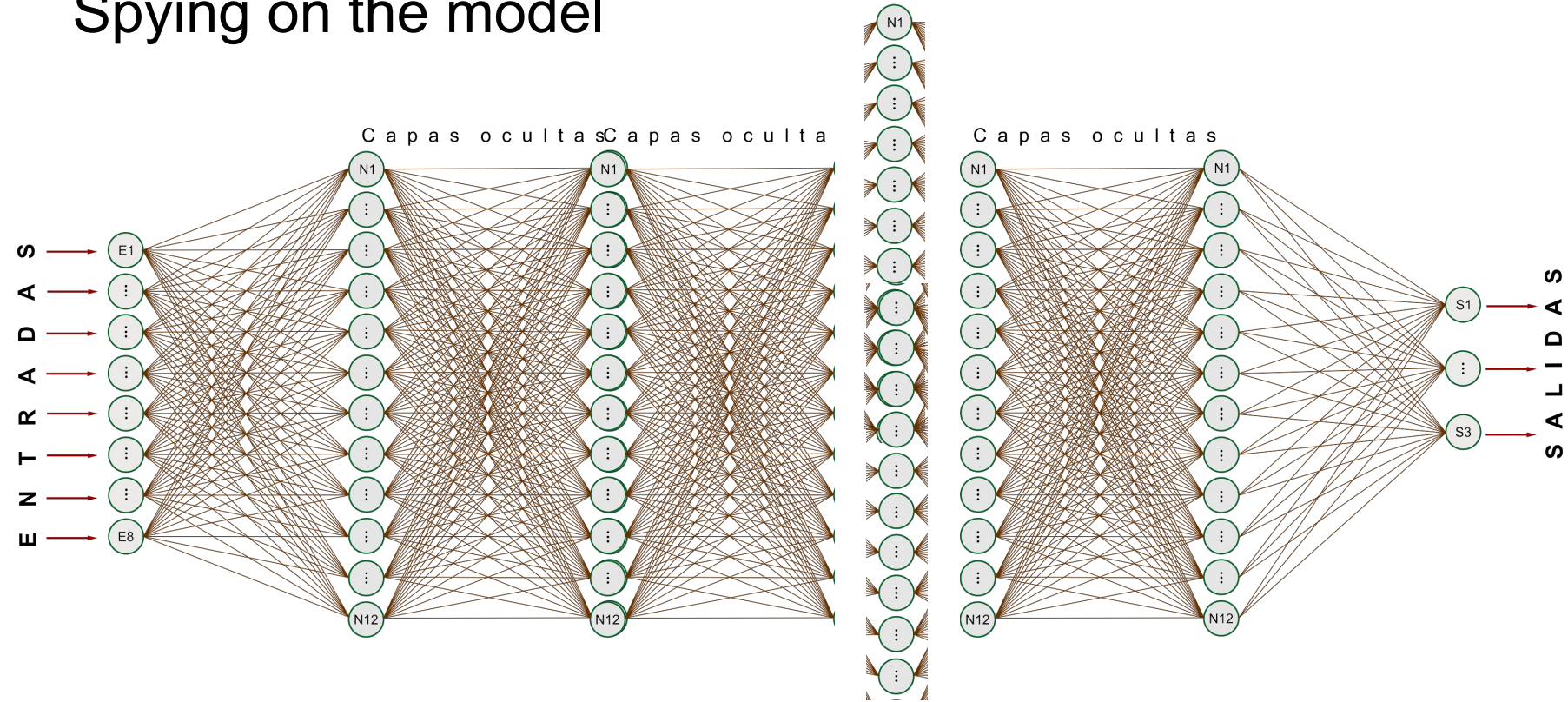




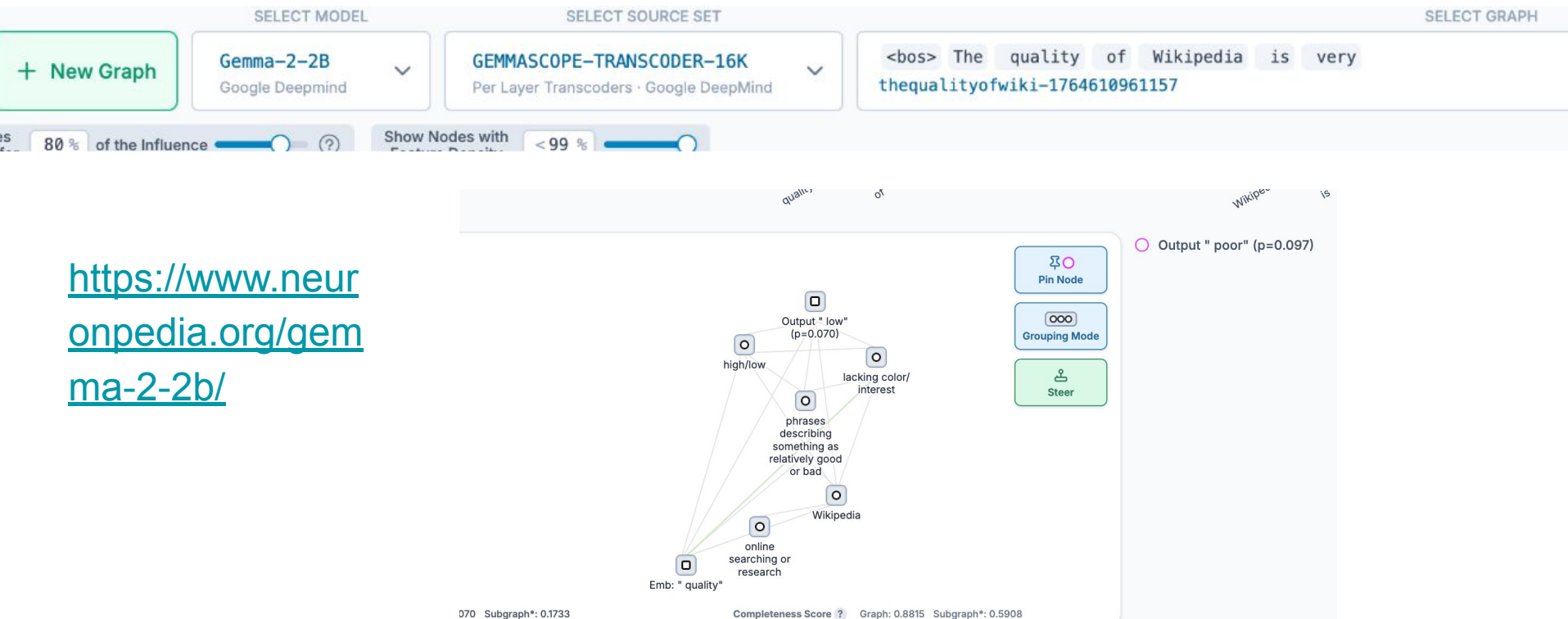
# Spying on the model



# Spying on the model



# Circuits



<https://www.neuronline.org/gemma-2-2b/>